

SCENARIO: BrightLayer Stores (simulated, 8 employees) uses: employee email accounts, shared computers, a company website, cloud storage with invoices, user passwords, and a public Wi-Fi router.

1. FIVE IMPORTANT ASSETS - list each asset and why it matters to the business.
 - a. Employee email accounts: Email accounts are used for personal communication, password reset, and business information.
 - b. Shared Computer: Employees can use those to access company's system, cloud storage. And no compromised system can affect company internally.
 - c. Company website: It provides information and services to customers.
 - d. Cloud Storage: It stores Financial and customer related records.
 - e. User passwords: It protect access to email, cloud storage, computers, and other systems.
2. THREE POTENTIAL THREATS - for each: who/what is the threat actor or event?
 - a. Phishing: Attackers could send fake emails to the company's employees and they could end up sharing company's vital informations.
 - b. Public wifi unauthorized access: An attacker could connect to company's public wifi and could harm the systems through network.
 - c. DDOS: An attacker could use many systems to send excessive traffic which will make it unavailable to the customers.
3. THREE VULNERABILITIES - for each: what is the exact weakness?
 - a. Shared System: Most employees use the same computer, making it difficult to control access or determine which user performed an action.
 - b. Weak or reused password: Employees might use simple passwords or the same password for different accounts. If one password is stolen, an attacker could use it to access other company accounts as well.
 - c. DDOS: Since it has a public website, it could be targeted by a DDoS attack. If there is no traffic filtering or DDOS protection, the website could become unavailable.
4. ASSOCIATED RISKS - link each threat+vulnerability pair to a potential loss (impact).
 - a. Phishing and weak passwords: An attacker could trick an employee into giving away their password. If the password is weak or reused, the attacker could gain access to the employee's email or cloud storage. This could expose invoices and other business information.
 - b. Malware with shared computers: If an employee accidentally downloads malware, it could infect a shared computer. This could lead to stolen or damaged files, loss of access to systems, and disruption to normal work.
 - c. DDOS: An attacker could send a huge amount of traffic to the company's website. Without proper protection, the website could go offline, preventing customers from accessing it. This could result in lost sales and damage to the company's reputation.
 - d.
5. SECURITY RECOMMENDATIONS - at least 5 improvements, each mapped to a risk above.
 - a. Use multi-factor authentication as this would reduce the chance of someone accessing an account even if they manage to steal a password.

- b. Give employees separate accounts on shared computers then this would make it easier to control access and identify which user performed an action.
- c. Install antivirus and keep software updated and this would help prevent malware from causing serious damage.
- d. Add DDoS protection to the company website so that this would help keep the website available during an attack.
- e. Train employees to recognize phishing emails. Employees should know how to identify suspicious links, attachments, and requests for passwords.
- f. Keep regular backups of important business data because If files are deleted, damaged, or encrypted by malware, the company would have a way to recover them.
- g. Secure the Wi-Fi router by changing the default administrator password, and using strong Wi-Fi security. This would reduce the chance of unauthorized access to the company's network.